Information Security

Unit 1

Introduction to Information Security

Foundations of security

- Security means protecting assets like data, systems, and networks from threats such as attackers, natural disasters, misuse, or loss.

- Information security focuses on protecting information and information systems from unauthorized access, use, disclosure, modification, disruption, or destruction.

- The dictionary defines security as the state of being free from danger and the measures taken to prevent crime, attack, espionage, or sabotage.

Foundations of security

- Physical security threats include theft of computers, damage due to fire, floods, or electrical surges, and loss of hardware or data.

- Rogue software such as viruses and malware are harmful programs that spread secretly and damage computer systems.

- Network security threats arise due to internet and network connectivity and include port scanning, spoofing, password cracking, spyware, and identity theft.

Computer Security Concepts

1. Origin of Information Security

- Information security started with computer security.
- Earlier, protection focused mainly on hardware and systems.

2. Types of Security

- **Network Security:** Protects networks, connections, and transmitted data from attacks.
- **Information Security:** Protects data's Confidentiality, Integrity, and Availability (CIA triad) whether stored, processed, or transmitted.
- **Physical Security:** Prevents unauthorized physical access to systems using locks, biometrics, surveillance, etc.

Computer Security Concepts

- **Data Security:** Focuses on protecting data from unauthorized access, disclosure, or modification.

3. Purpose of Security

- Allows safe collaboration while managing risk.
- Balances data availability with confidentiality.
- Required because modern systems rely heavily on digital data processing.

4. Traditional Security Methods

Earlier, organizations relied mainly on:

- Physical protection (guards, locks)
- Administrative controls (policies, permissions)

Now, technical security controls are also essential.

Computer Security Concepts

5. Need for Network Security

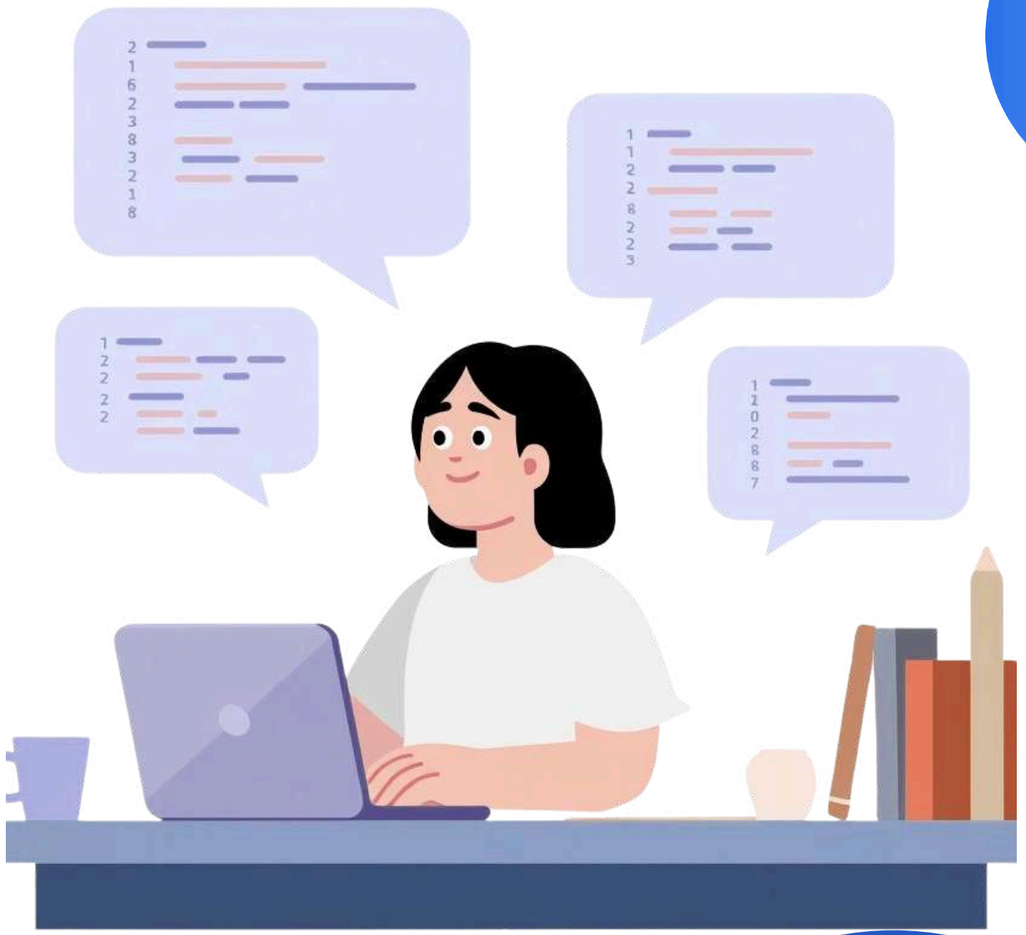
- Data transmitted over networks can be intercepted or altered.
- Security mechanisms are required to protect information during transmission.

6. Examples of Security Violations

- **Eavesdropping:** Unauthorized user captures data while it is being transmitted.
- **Repudiation:** Sender denies sending a message after consequences occur.
- **Message Tampering:** Attacker intercepts and modifies message content before forwarding.

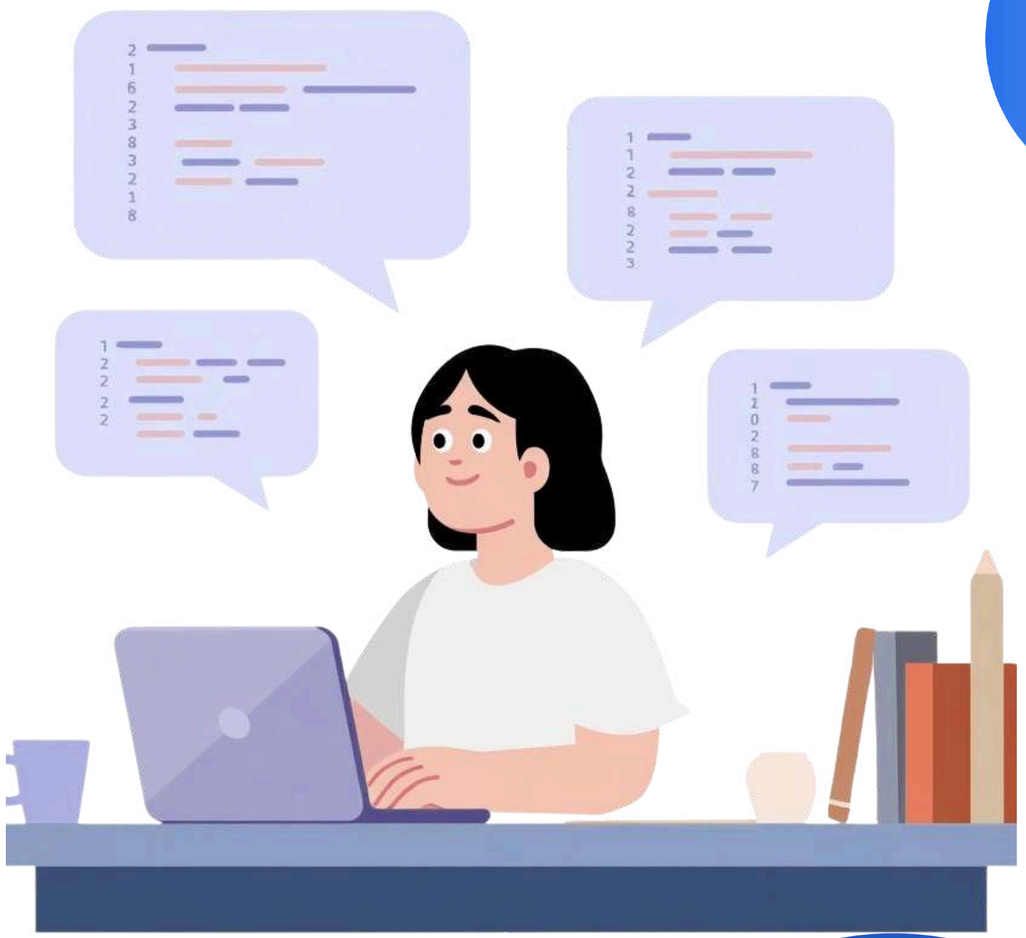
Basic Terminology in Security

- **Cryptography:** Science of converting readable message (plaintext) into unreadable form (ciphertext) and back.
- **Plaintext:** Original readable message.
- **Ciphertext:** Encrypted unreadable output depending on plaintext + key
- **Cipher:** Algorithm used to transform plaintext into ciphertext using substitution or transposition.
- **Key:** Secret value used in encryption and decryption known only to sender and receiver.



Basic Terminology in Security

- **Encryption:** Process of converting plaintext → ciphertext using cipher + key.
- **Decryption:** Process of converting ciphertext → plaintext using cipher + key.
- **Cryptanalysis:** Technique of breaking encrypted message without knowing key (code-breaking).
- **Cryptology:** Combination of cryptography + cryptanalysis.
- **Code:** Method of converting message into another form using a codebook.



Categories of Computer Security

1.Cryptography

- Technique of encrypting and decrypting data
- Converts readable data into unreadable form for protection

2.Data Security

- Protects data from unauthorized access, modification, or corruption
- Ensures data integrity and confidentiality

3.Computer Security

- Protects computer systems and resources from attacks and misuse
- Includes prevention, detection, and protection mechanisms

Categories of Computer Security

4.Network Security

- Protects data during transmission over networks
- Prevents interception, unauthorized access, and data theft

Security Techniques

1.Encryption

- Protects information by converting readable data into unreadable form using cryptographic techniques. Different encryption types provide different levels of security.

2.Access Control

- Security mechanism that regulates who or what can view or use resources in a computing environment. Ensures only authorized users get access.

3.Data Backup

- Process of storing additional copies of data in separate physical or virtual locations. Helps recover data if original data is lost, corrupted, or deleted.

Security Techniques

4.Firewall

- Network security system that monitors incoming and outgoing traffic and allows or blocks data based on predefined security rules.

5.Antivirus Software

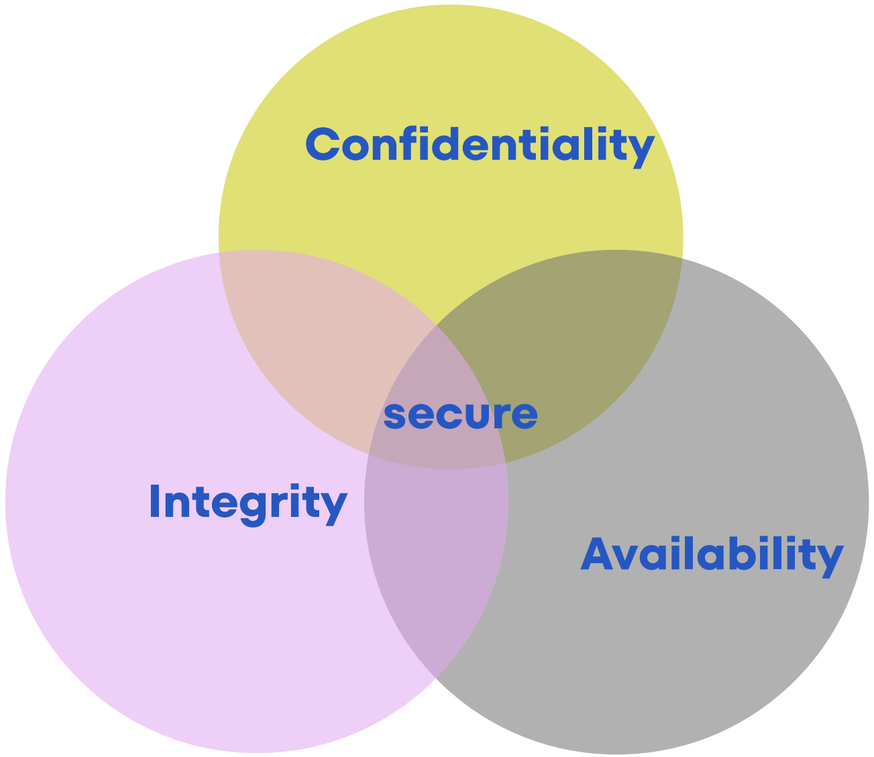
- Security program designed to detect, prevent, and remove malware and other threats. Provides real-time protection and system scanning.

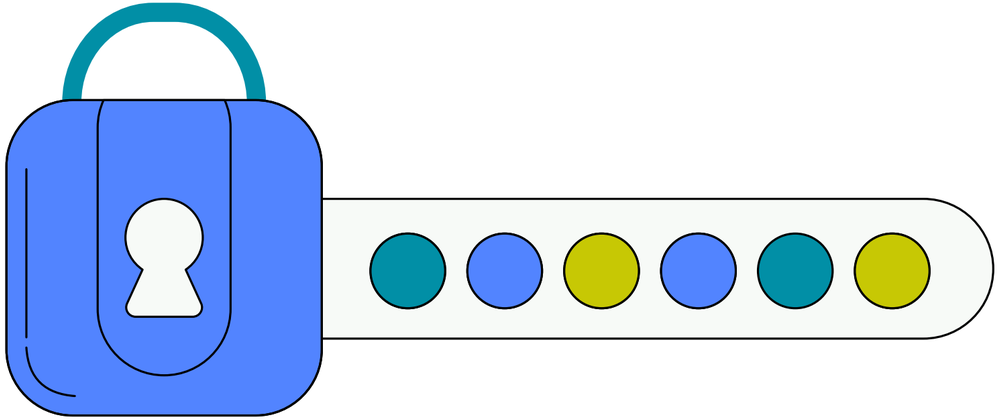
6.Intrusion Detection System (IDS)

- Monitors network or system activities for malicious actions or policy violations and alerts administrators.

Elements of Information Security

CIA Triad

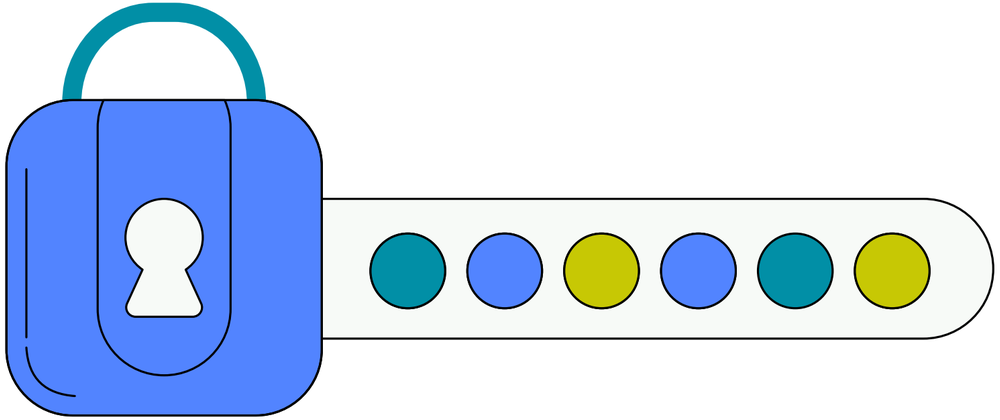




Elements of Information Security

Confidentiality

- Limits access to information only to authorized users
- Prevents unauthorized disclosure of data
- Sensitive information must remain secret from unauthorized individuals



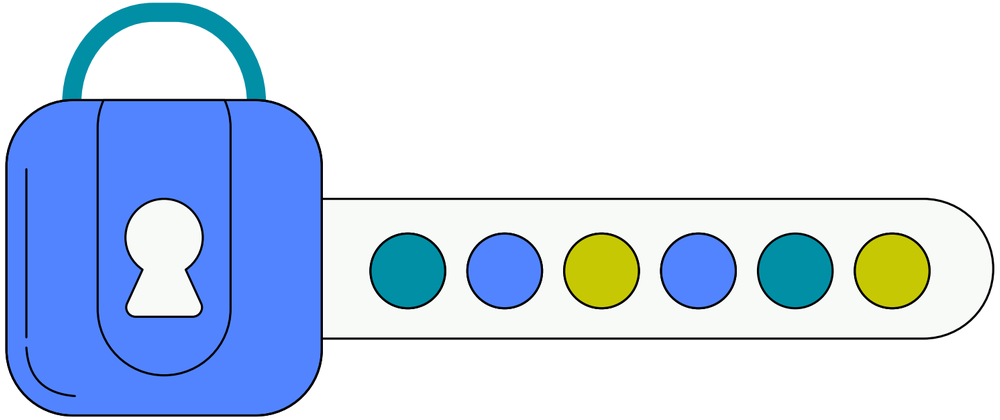
Integrity

- Ensures trustworthiness and accuracy of information
- Data should not be altered without detection
- Includes concept of data integrity (data should not be modified accidentally or maliciously)

Elements of Information Security

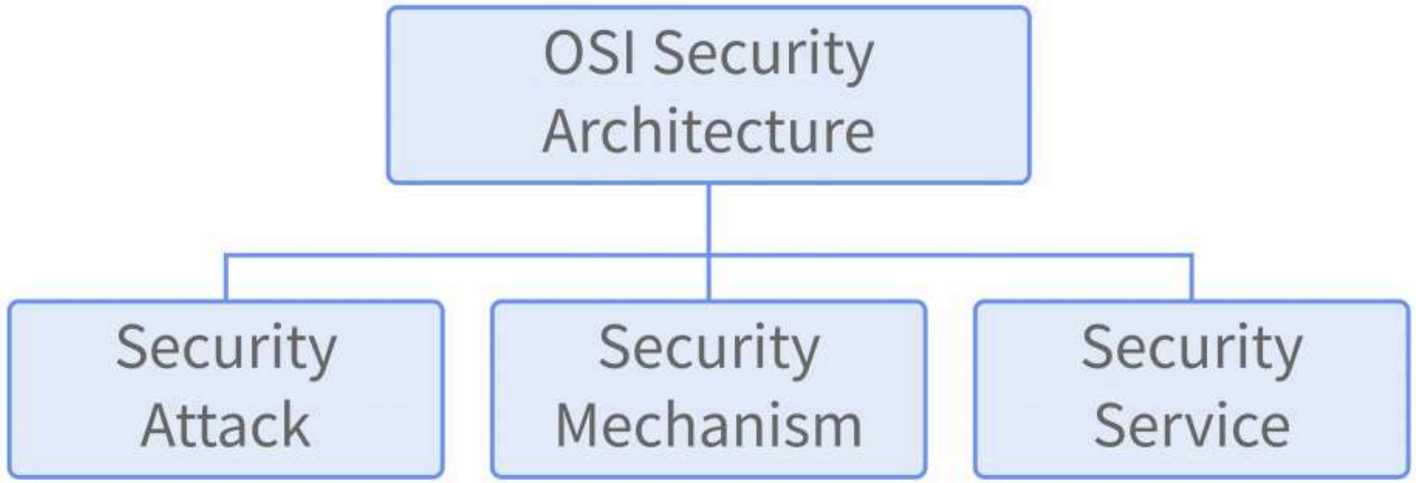
Availability

- Ensures information and systems are accessible when needed
- Authorized users must not be prevented from accessing resources
- System downtime can be as harmful as data loss
- Availability importance increases when organizations depend heavily on IT infrastructure



OSI Security Architecture

- Defines a structured approach for implementing security at each network layer
- Focuses on three main concepts:
- Security Attacks, Security Mechanisms, Security Services



OSI Security Architecture

Security Attacks

- Attempt by an unauthorized entity to access, damage, or misuse systems or data

Two main types:

- **Passive Attacks:** Monitor data without altering it (e.g., eavesdropping, traffic analysis)
- **Active Attacks:** Modify or disrupt data/system
- Masquerade
- Replay attack
- Message modification
- Denial of Service (DoS)

OSI Security Architecture

Security Mechanisms

- Techniques designed to detect, prevent, or recover from security attacks
- **Examples:**
  - Encryption
  - Digital signatures
  - Authentication mechanisms

Security Services

- Services that enhance system security and protect data
- Types include:
  - Authentication
  - Access control

Information Security Attacks

- A security attack is an unauthorized attempt to steal, damage, modify, or expose data from an information system.

- Attacks are mainly classified into two types: Passive attacks and Active attacks.

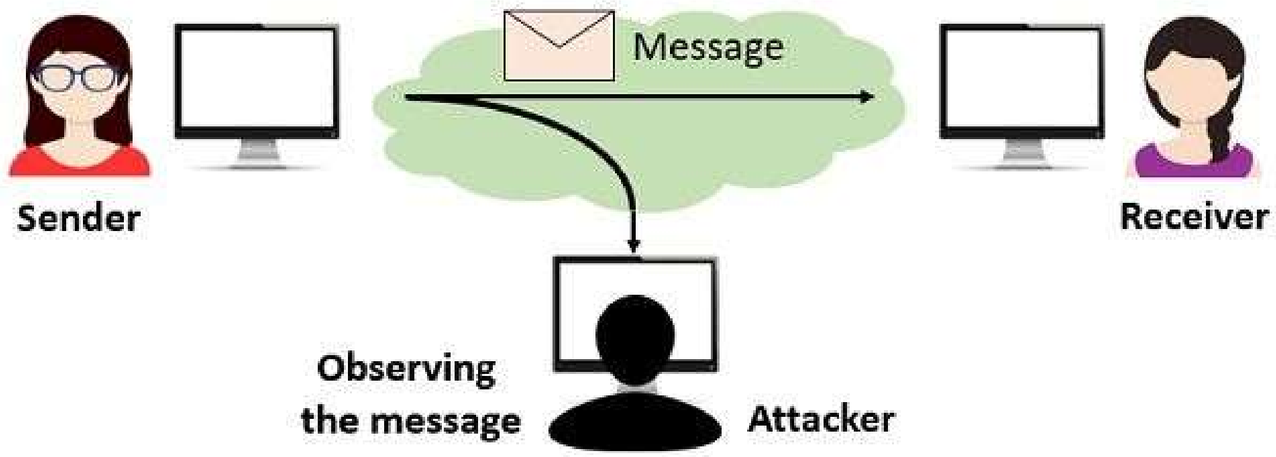
Passive Attacks

- Attacker monitors or listens to data transmissions without altering them.
- Goal is to obtain information secretly.
- Does not affect system resources directly.

Information Security Attacks

Types:

- **Release of Message Content:** Unauthorized reading of emails, calls, or files containing confidential data.
- **Traffic Analysis:** Even if message content is encrypted, attacker studies communication patterns to gather information.



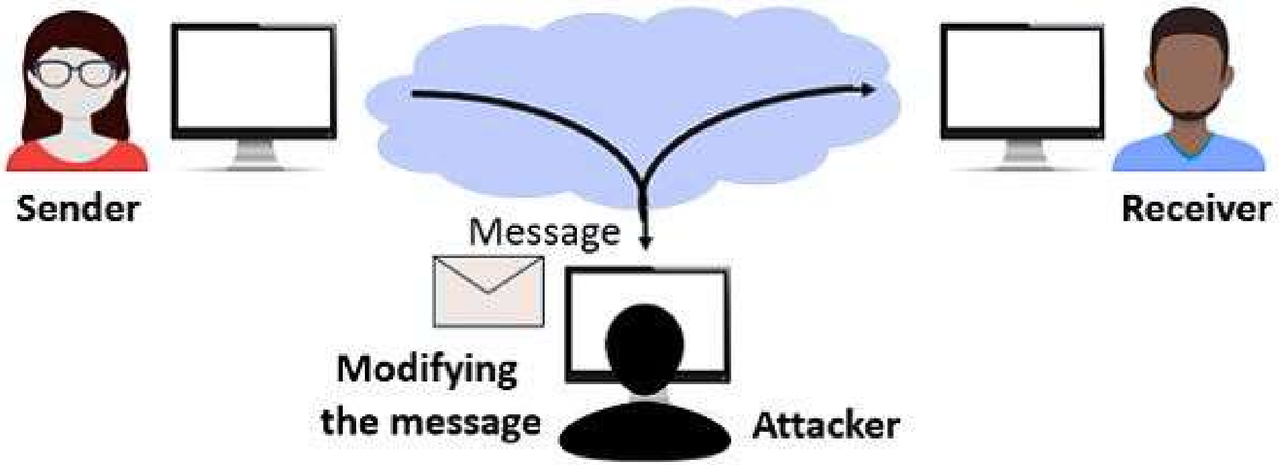
Information Security Attacks

Active Attacks

- Involve modification of data or creation of false data streams.
- More dangerous and harder to prevent.
- **Types:**
  - **Masquerade:** Attacker pretends to be another user or entity.
  - **Replay Attack:** Captured data is retransmitted to gain unauthorized access.
  - **Modification of Messages:** Data is altered during transmission.
  - **Denial of Service (DoS):** Attacker overloads system or network to make services unavailable.

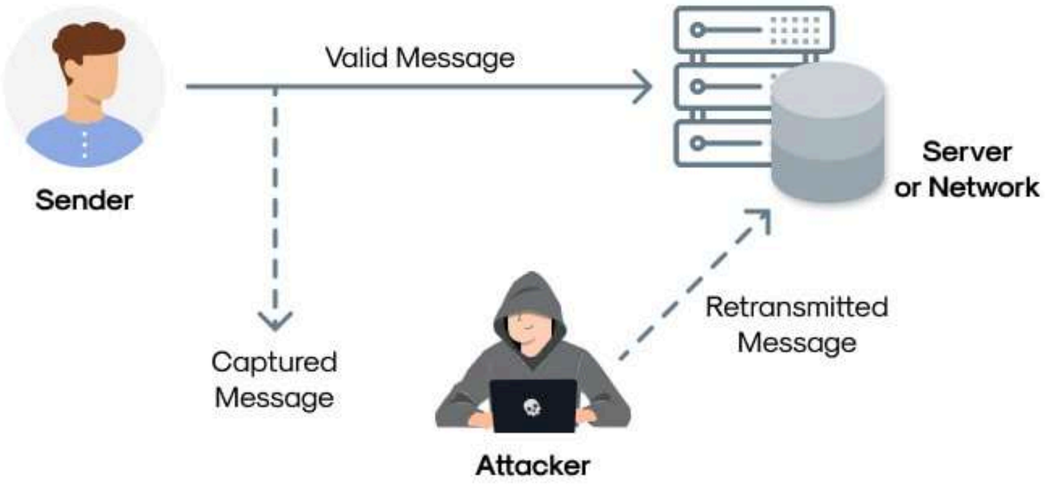
Information Security Attacks

Active attack



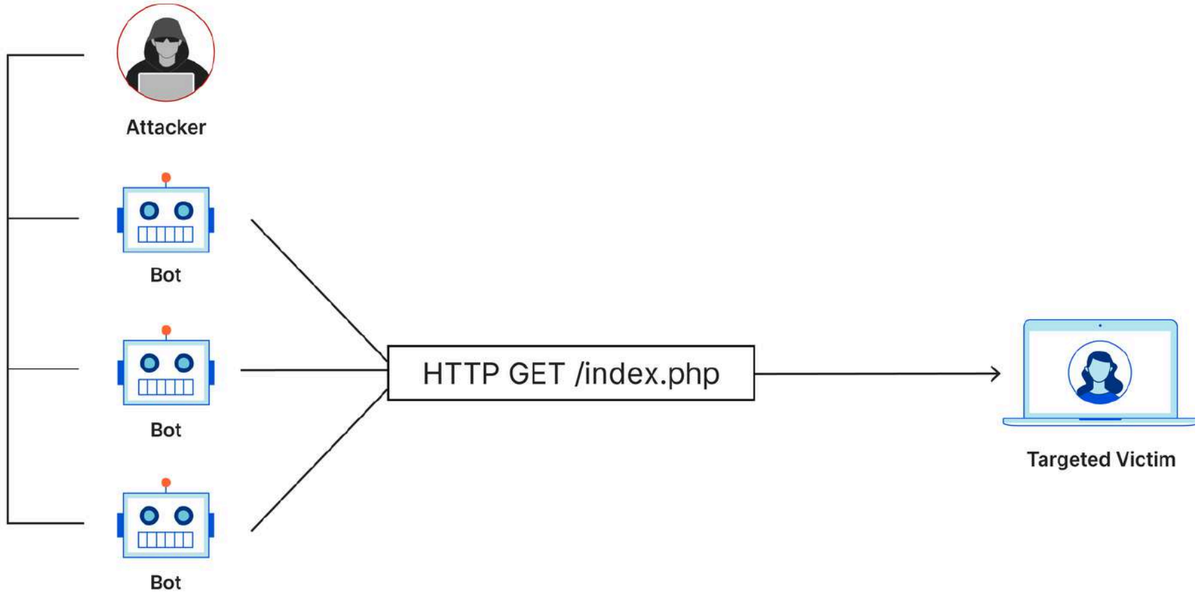
Information Security Attacks

Replay attack



Information Security Attacks

Denial of service attack



Information Security Attacks

Difference

Active Attack	Passive Attack
Involves modification of data or system	Only monitors or observes data
Attacker alters, deletes, or injects data	Attacker only reads or listens
Affects system resources	Does not affect system resources
Easier to detect	Hard to detect
Threatens integrity & availability	Threatens confidentiality
Examples: DoS, message modification, masquerade, replay	Examples: eavesdropping, traffic analysis

Security Services

- Security services are communication or processing functions that provide protection to systems and implement security policies using security mechanisms.

Authentication

- Verifies identity of communicating entity
- Ensures sender/receiver is genuine
- Common method: login passwords
- Types:
  - Peer entity authentication
  - Data origin authentication

Security Services

Access Control

- Determines who can access information or system resources
- Prevents unauthorized use of data or systems

Data Confidentiality

- Protects data from unauthorized disclosure
- Ensures only authorized users can view information

Non-Repudiation

- Prevents denial of sending or receiving a message
- Sender cannot deny sending message
- Receiver cannot deny receiving message

Security Mechanisms

- Security mechanisms are methods designed to detect, prevent, or recover from security attacks and protect system resources.

Encipherment (Encryption)

- Hides data by converting it into unreadable form
- Achieved using mathematical algorithms
- Based on cryptography techniques

Digital Signature

- Used for authentication and verification
- Sender signs message using private key
- Receiver verifies using sender's public key

Security Mechanisms

Access Control

- Restricts unauthorized users from accessing resources
- Ensures only permitted users can use system/data

Authentication Exchange

- Verifies users by exchanging authentication information
- Confirms identity before communication

Traffic Padding

- Adds extra bits or data to messages
- Prevents attackers from analyzing traffic patterns

Security Policy

- A security policy is a formal document that defines rules, expectations, and procedures an organization follows to maintain security.
- It guides how systems, data, and resources should be protected.

Types / Areas Covered

- Controls security risks and threats
- Prevents identity theft
- Helps detect system misuse
- Supports fraud prevention and monitoring

Security Policy

Basic Privacy Policies

- Set of standards and guidelines for protecting user data
- May follow legal or regulatory requirements
- Defines acceptable and unacceptable actions

Advisory Policies

- Written as rules, do’s and don’ts for users
- Guide safe behavior while using systems

User Policy

- Specific rules related to individual users
- Defines responsibilities and permitted actions

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